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SILKIE: PREFERENCE DISTILLATION FOR LARGE VISUAL LANGUAGE MODELS

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ABSTRACT

This paper explores preference distillation for large vision language models (LVLMs), improving their ability to generate helpful and faithful responses anchoring the visual context. We first build a vision-language feedback (VLFeedback) dataset utilizing AI annotation. Specifically, responses are generated by models sampled from 12 LVLMs, conditioned on multi-modal instructions sourced from various datasets. We adopt GPT-4V to assess the generated outputs regarding helpfulness, visual faithfulness, and ethical considerations. Furthermore, the preference supervision is distilled into Qwen-VL-Chat through the direct preference optimization (DPO) method. The resulting model Silkie, achieves 6.9% and 9.5% relative improvement on the MME benchmark regarding the perception and cognition capabilities, respectively. Silkie also demonstrates reduced hallucination by setting a new state-of-the-art score of 3.02 on the MMHal-Bench benchmark. Further analysis shows that DPO with our VLFeedback dataset mainly boosts the fine-grained perception and complex cognition abilities of LVLMs, leading to more comprehensive improvements compared to human-annotated preference datasets. Project page: https://vlf-silkie.github.io.

1 INTRODUCTION

In recent years, there has been a rapid evolution in the development of large vision language models (LVLMs), epitomized by the remarkable achievements of the GPT-4V model (OpenAI, 2023a). The integration of language and visual components through modality alignment (Alayrac et al., 2022; Awadalla et al., 2023), coupled with supervised fine-tuning on instruction tuning datasets (Liu et al., 2023c; Li et al., 2023c; Zhu et al., 2023), has endowed LVLMs with promising perceptual and cognitive abilities (Yang et al., 2023). This augmentation enables them to engage in nuanced reasoning over images and deliver responses according to user queries. Despite these advancements, open-sourced LVLMs still exhibit misalignment issues that pose a potential risk of generating misleading content without anchoring to the provided visual context (Li et al., 2023d; Sun et al., 2023), or biased responses against minority groups (OpenAI, 2023a). Consequently, addressing and rectifying the misalignment behavior of LVLMs is crucial to enhance their reliability and mitigate the risk of generating biased outputs.

In this paper, inspired by the success of reinforcement learning from AI feedback (RLAIF) in improving LLMs alignment with AI preference (Tunstall et al., 2023; Lee et al., 2023; Cui et al., 2023), we enhance LVLMs by distilling the preference annotation from a powerful LVLM, i.e., GPT-4V (OpenAI, 2023a). We first construct a high-quality instruction set by collecting from various multi-modal instruction tuning sources, including general conversation and reasoning instructions from LLaVA (Liu et al., 2023c) and SVIT (Zhao et al., 2023a), academic task instructions from M3IT (Li et al., 2023c), special datasets such as medical image understanding from PMC-VQA (Zhang et al.,

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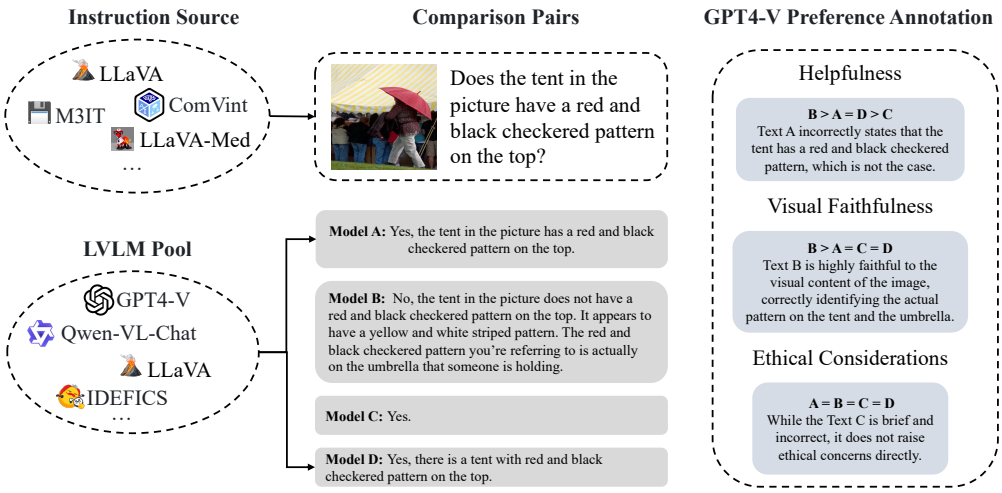

Figure 1: The annotation framework of our vision-language feedback (VLFeedback) dataset. We collect instructions from various sources and decode the corresponding responses using 4 models sampled from the LVLM pool. The GPT-4V model assesses these responses regarding three aspects, providing ratings and rationales for the scores.

2023a) and text-rich understanding from LLaVaR and complex visual reasoning from ComVint (Du et al., 2023). There are 80k instructions in total, covering 65k unique images. We further built a pool of 12 LVLMs including BLIP-family Li et al. (2023b); Dai et al. (2023), LLaVA-series (Liu et al., 2023c;b; Sun et al., 2023), Fuyu-8B (Bavishi et al., 2023), Qwen-VL (Bai et al., 2023) and GPT-4V (OpenAI, 2023a), to obtain corresponding responses conditioned on our collected instructions. We randomly sample 4 models for each instruction to obtain the corresponding outputs, resulting in 28k multi-modal instruction and response pairs for preference annotation. As annotating response preferences with human annotators is infeasible regarding large-scale responses and can be cumbersome due to the subjectivity in the annotation process, GPT-4V is adopted to assess the quality of different model outputs. To achieve this, we define evaluation templates to evaluate LVLMs outputs from three aspects: (i) *Helpfulness*, aims to evaluate whether the response is relevant for the user query and improves the user understanding for the visual content; (ii) *Visual Faithfulness*, focuses on the consistency between the visual clues and response, aiming for detecting potential ungrounded hallucination; and (iii) *Ethical Considerations*, targets to examine whether the response contains any offensive or biased content. GPT-4V is then queried with the annotation templates, the image, and corresponding instructions to assess the quality of different model outputs. The annotated preference ranking dataset, named vision-language feedback (VLFeedback), enables us to explore the effect of preference distillation for LVLMs.

With the VLFeedback dataset, we use direct preference optimization (DPO) (Rafailov et al., 2023a) to build our Silkie model based on Qwen-VL-Chat (Bai et al., 2023). Our experimental results reveal that the preference distillation effectively improves the overall capabilities of multi-modal tasks. For example, on the MME benchmark (Fu et al., 2023), Silkie outperforms the base model by 6.9% and 9.5% in perception and cognition tasks, respectively. Furthermore, with visual faithfulness as an annotation aspect, Silkie could produce responses that are more aligned with the visual context, indicated by the new state-of-the-art score of 3.02 on the hallucination evaluation benchmark MMHal-Bench (Sun et al., 2023). Additional in-depth investigation of performance improvements shows that VLFeedback provides more pronounced improvements on fine-grained perception tasks such as Optical Character Recognition (OCR) and complex cognition-level reasoning tasks such as code reasoning. Intriguingly, we also find that the AI-annotated preference boosts the LVLMs more consistently than the human-annotated preference dataset (Yu et al., 2023), potentially due to the comprehensive coverage of our preference dataset. These findings substantiate the value of our VLFeedback dataset for future studies towards building LVLMs better aligned with humans.

In summary, the contribution of this work is two-fold: (i) We construct VLFeedback, a large-scale multi-modal preference dataset annotated by GPT-4V on three curated aspects, covering 80k multi-

Dataset	Dataset Description	# Sampled Instructions
LLaVA	Visual Instruction Synthesized by GPT-4	19,614
SVIT	Scaled-up Visual Instruction Synthesized by GPT-4	22,823
LLaVAR	Text-rich Image Understanding	13,770
LRV	Robust Visual Instruction	12,357
LLaVAMed	Biomedical Vision-Language Instruction	5,861
ComVint	Complex Visual Reasoning Instruction	2,384
PMC-VQA	Medical Image Question Answering	2,364
M3IT	Academic Vision-Language Tasks	687
PCA-EVAL	Embodied Decision-making Instruction	398
Total	Visual instruction in multi-domains	80,258

Table 1: Descriptions and statistics of multi-modal instructions in our VLFeedback dataset.

modal instructions and decoded by 12 performant LVLMs; (ii) Experimental results and analysis demonstrate that performing DPO on our VLFeedback dataset improves LVLMs comprehensively, providing insights for future LVLM alignment research.

2 Visual-Language Feedback Dataset

In this section, we elaborate on the construction process of our visual-language feedback (VLFeedback) dataset, as illustrated in the Figure 1. We first introduce the multi-modal instructions sources (§2.1), followed by the details of selected LVLMs for decoding (§2.2) and the annotation with GPT-4V (§2.3). Finally, we present the statistics of our VLFeedback dataset (§2.4).

2.1 Instruction Source

We curate instruction sources from diverse datasets that span various capabilities of LVLMs across different domains. Our selection encompasses:

- **General Vision-Language Instructions:** Featuring datasets such as LLaVA (Liu et al., 2023c) and SVIT (Zhao et al., 2023a), these datasets are constructed by inputting textual descriptions of images to ChatGPT/GPT-4. They prompt the generation of visual-related instructions that encompass diverse types, including detailed descriptions, reasoning processes, and interactive conversations.
- **Academic Vision-Language Instructions:** Drawn from 20 samples of each task in M3IT (Li et al., 2023c), this set offers comprehensive coverage of previous academic vision-language tasks such as visual question answering, image captioning and classification.
- **Robustness-oriented Vision-Language Instructions:** Challenging instructions from datasets like LRV (Liu et al., 2023a), demanding complex visual reasoning from LVLMs, and ComVint (Du et al., 2023), which introduces misleading queries in the instructions, are incorporated to enrich the coverage of our dataset.
- **Domain-specific Vision-Language Instructions:** We incorporate LLaVAR (Zhang et al., 2023b), emphasizing text-rich images like documents and logos; PMC-VQA (Zhang et al., 2023a) for medical images; LLaVAMed (Li et al., 2023a) for biomedical images; and PCA-EVAL (Chen et al., 2023a), designed for visual decision-making instructions in embodied environments. These instructions require domain knowledge that is potentially useful for downstream applications.

Table 1 summarizes the characteristics and statistics of instruction sources sampled in our paper.

2.2 Model Pool

We have curated a diverse model pool comprising 12 LVLMs to cover recent advancements, including:

Visual Faithfulness Assessment
Definition: Evaluate whether the generated response is aligned with the image content, avoiding ungrounded statements.

Guidelines:
- Ensure that the generated response accurately reflects the visual elements present in the image.
- Flag instances where the model provides ungrounded statements that do not align with the content of the image.
- Assess the level of consistency between the generated text and the visual information.

Scoring: Rate outputs 1 to 5 based on the following criteria:
1. Significantly Inaccurate: The response is significantly inaccurate and does not align with the image content.
2. Some Inaccuracy / Minor Deviations: The response contains some inaccuracies or minor deviations from the image content.
3. Moderately Faithful: The response is moderately faithful but may have subtle inaccuracies.
4. Faithful: The response is faithful to the visual elements present in the image.
5. Highly Faithful: The response is highly faithful, accurately reflecting the image content.

Table 2: Visual faithfulness assessment annotation guideline for GPT-4V model.

- **GPT-4V** (OpenAI, 2023a), the proprietary vision language models developed by OpenAI, which are shown to be powerful on various multi-modal tasks (Yang et al., 2023).
- **LLaVA-series models**, which adopt Vicuna models as the backbone and are trained on the GPT-4 (text-only) synthesized multi-modal dataset. We select the enhanced version `LLaVA-v1.5-7B` and `LLaVA-v1.5-13B` (Liu et al., 2023b), and the RLHF version with visual faithfulness alignment, LLaVA-RLHF (Sun et al., 2023) with different image resolutions `LLaVA-RLHF-7b-v1.5-224` and `LLaVA-RLHF-13b-v1.5-336`.
- **Qwen-VL-Chat** (Bai et al., 2023), which show promising capabilities on various vision-language benchmarks with scaled-up multi-modal pre-training and supervised fine-tuning on curated datasets.
- **IDEFICS-9b-Instruct** (Laurençon et al., 2023), which is a open-sourced implementation of Flamingo (Alayrac et al., 2022), supporting interleaved image-text inputs. After training on publicly available image-text alignment pairs and instruction tuning datasets, it demonstrates comparable results with the original closed-source model on various image-text benchmarks.
- **Fuyu-8B** (Bavishi et al., 2023), which introduces a novel architecture by segmenting images into patches and training a conditional language model from scratch, showcasing the great potential to deal with high-resolution images.
- **InstructBLIP** (Dai et al., 2023), which employs an instruction-aware visual feature extraction module based on BLIP2 (Li et al., 2023b). We select `InstructBLIP-Vicuna-7B` and `InstructBLIP-Vicuna-13B` with different LLMs as the backbone models.
- **VisualGLM-6B** (Du et al., 2022) is an open-sourced, multi-modal dialog language model supporting images, Chinese, and English.
- **MM-ICL** (Zhao et al., 2023b), which is built on BLIP2 (Li et al., 2023b) and has been further enhanced via training on a curated interleaved image-text dataset to enhance the in-context learning ability. We adopt `MMICL-Vicuna-13B` for decoding.

For each instruction, we randomly sample four models for decoding. The decoding hyper-parameters adhere to the recommendations provided in the original implementations.

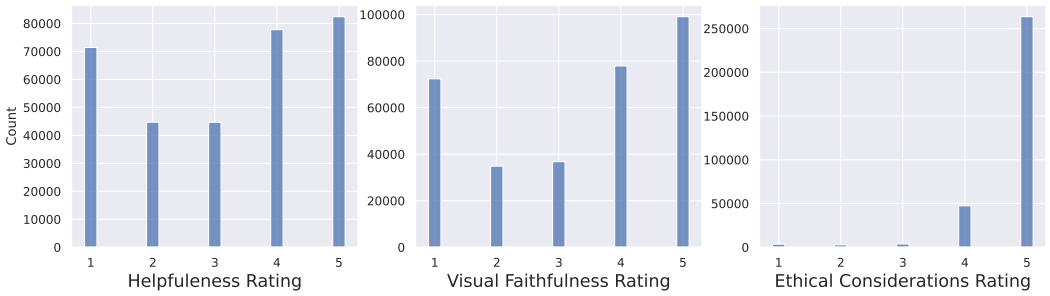

Figure 2: Rating distribution of different aspects. Helpfulness and Visual Faithfulness share similar score distributions, with the majority of decoded responses evaluated without Ethical Considerations.

2.3 GPT-4V Aided Preference Annotation

Inspired by the recent progress in alignment from AI Feedback (Bai et al., 2022b; Lee et al., 2023; Cui et al., 2023), we define *Helpfulness* for judging whether the response is relevant and helps the user, and *Ethical Considerations* to avoid potential inappropriate responses that may contain toxic content such as biases or violence. Furthermore, considering the characteristics of LVLMs involving the interaction between modalities, we design a special *Visual Faithfulness* criterion to evaluate the response consistency between modalities. Specifically, we ask the GPT-4V model to assess the response quality given the original image and instruction, rating the visual faithfulness from 1 to 5. The annotation template for visual faithfulness can be found in Table 2, and we include the annotation templates for helpfulness and harmlessness in Appendix A.

2.4 Preference Statistics

We present statistics on the annotated results to elucidate the distribution of the annotation scores.

Score Distribution in Different Aspects In Figure 2, we illustrate the score distributions for three distinct aspects. (1) Helpfulness: The majority of samples garnered scores exceeding 4, while a notable portion of samples received the lowest score. This suggests the general effectiveness of LVLMs in meeting the intended objectives of the annotations, indicating the successfully performed instruction tuning. (2) Visual Faithfulness: Scores for visual faithfulness closely mirror the distribution observed in the helpfulness evaluation, implying a potential correlation between these two aspects during the annotation process. The similarity in distributions suggests that the perceived helpfulness of the content likely influences judgments on visual faithfulness. (3) Ethical Considerations: Interestingly, only a limited portion of the annotated instructions exhibit potential ethical considerations. This observation may be attributed to the predominant nature of the sampled instructions, which may not be primarily geared toward red-teaming prompts (Perez et al., 2022) designed to elicit harmful results from the LVLMs. Notably, this finding prompts consideration for a more targeted preference annotation focused explicitly on ethical considerations in future endeavors.

Score Differences between Models Table 3 lists the scores of different models regarding three aspects. As the evaluated LVLMs may adopt the annotated instructions as the training data, we would like to note that this score comparison could be unfair for certain models. Nevertheless, GPT-4V demonstrates a clear advantage over open-sourced LVLMs, showcasing its great potential to serve as a proxy for human annotators to provide feedback. We further select two representative models, GPT-4V and Qwen-VL-Chat, to delve into the distribution of annotated scores. Figure 3 depicts the distinctions between these models. Notably, GPT-4V consistently obtains higher ratings across all three facets, evidenced by a prevalence of samples with scores equal to or greater than 4, echoing the results in the average ratings. It is important to acknowledge that GPT-4V's dominance may stem from its role as the annotator, introducing a potential bias towards its own characteristics and proclivity for detailed responses. Despite this, Qwen-VL-Chat still exhibits better results in the helpfulness and visual faithfulness evaluation than in the overall performance of all models as

Model	Helpfulness	Visual Faithfulness	Ethical Considerations	Average
GPT-4V	4.54	4.59	4.96	4.70
LLaVA-1.5-13B	3.47	3.63	4.84	3.98
Qwen-VL-Chat	3.33	3.62	4.86	3.94
LLaVA-1.5-7B	3.40	3.54	4.83	3.92
LLaVA-RLHF-13b-v1.5-336	3.49	3.40	4.75	3.88
IDEFICS-9B-Instruct	3.12	3.40	4.90	3.81
LLaVA-RLHF-7b-v1.5-224	3.35	3.27	4.75	3.79
InstructBLIP-Vicuna-7B	2.71	2.96	4.80	3.49
InstructBLIP-Vicuna-13B	2.71	2.94	4.80	3.49
Fuyu-8B	2.53	2.82	4.77	3.37
VisualGLM-6B	2.24	2.27	4.55	3.02
MMICL-Vicuna-13B	1.51	1.51	4.01	2.34

Table 3: Average score in three aspects and the overall performance. GPT-4V shows an evident advantage over open-sourced LVLMs, motivating us to adopt GPT-4V as a proxy of human annotators.

presented in Figure 2. This suggests Qwen-VL-Chat's commendable competence in addressing diverse user queries, motivating us to adopt it as a backbone model for future explorations.

Preference Agreement between GPT-4V and Human Annotators Given that the efficacy of RLHF hinges on accurately rated human preferences and the AI evaluator can become unstable (Wang et al., 2023), we undertake a validation experiment by calculating the agreement rate between human annotators and GPT-4V. We asked three human annotators to compare the overall quality of two responses given the same annotation guide for GPT-4V. The experiment is conducted on a subset of 100 randomly sampled comparisons from our VLFeedback dataset, revealing an impressive average agreement rate of 83.1%. This finding further underscores the reliability of employing GPT-4V for annotating preference data, substantiating its credibility in this crucial role.[1]

3 Preference Distillation for LVLMs

Previous results have shown that performant open-sourced LVLMs have been equipped with promising abilities after sufficient instruction tuning. Therefore, in this work, we explore whether learning from the preference data can improve LVLMs regarding helpfulness and visual faithfulness. Our method builds upon the VLFeedback dataset and distills vision-language AI preferences with direct preference optimization (DPO) (Rafailov et al., 2023b).

Task Formulation Let x be a prompt containing both images and text inputs, and y_i denotes the corresponding response generated by model π_i, with scores annotated by GPT-4V in three aspects: s_i^h for helpfulness, s_i^v for visual faithfulness and s_i^e for ethical consideration, respectively. To fully utilize the fine-grained annotations in various aspects, we average the scores of three aspects into an overall rating s_i to compare model responses for the same prompt, resulting in an ordered list of responses $\{y_1, \ldots, y_K\}$. Following InstructGPT (Ouyang et al., 2022), the list of K responses is then mapped into $K(K-1)/2$ comparisons. Pairs with tied scores are disregarded. The final preference dataset $\mathcal{D}$ used for fine-tuning consists of triples of one prompt and two responses (x, y_w, y_l), where y_w is the chosen response with a higher score and y_l is the response labeled as rejected.

Preference Alignment Optimization To align models with preference data, the prevalent RLHF pipeline is to optimize the following objective (Stiennon et al., 2020a):

$$\max_{\pi_\theta} \mathbb{E}_{x\sim\mathcal{D}, y\sim\pi_\theta(y|x)} \left[r(x,y)\right] - \beta \mathbb{D}_{\mathrm{KL}} \left[\pi_\theta(y \mid x) \| \pi_{\mathrm{ref}}(y \mid x)\right],$$

where r is the reward model and the KL term penalizes deviations of the current model π_θ from the initial model π_{ref}. This optimization can be done in a two-stage manner, by first learning a reward model $r_\phi(x, y)$ on comparison pairs under the Bradley-Terry (BT) model (Bradley & Terry,

[1] In Appendix C, we provide examples of human-GPT disagreements, demonstrating instances where GPT-4V generates incorrect annotations due to misjudgment regarding visual contents or conflicting rationales.

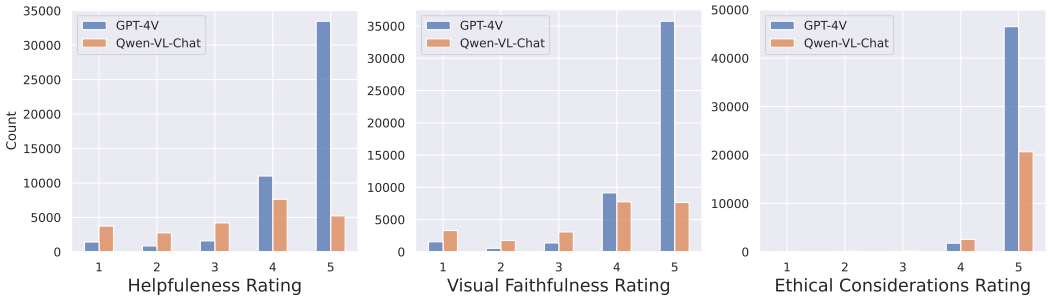

Figure 3: Score distribution comparison between GPT-4V and Qwen-VL-Chat.

1952) and then using online RL algorithms (e.g., proximal policy optimization (PPO) (Schulman et al., 2017b)) to optimize the model with respect to rewards. However, this approach necessitates an additional reward model and iterating fine-tuning the model and extensive sampling, leading to training instability and high computational cost. Direct preference optimization (DPO) mitigates these issues by directly fine-tuning the model on the preference data, bypassing the reward modeling stage. The key insight behind DPO is that the optimal policy π^* has a closed-form solution with regard to a reward function r and initial policy π_{ref}:

$$r(x, y) = \beta \frac{\pi^*(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \beta \log Z(x),$$

where Z is the partition function. Under the BT preference model, the objective becomes:

$$\max_{\pi_\theta} \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \log \sigma \left(\beta \log \frac{\pi_\theta\left(y_w \mid x\right)}{\pi_{\text{ref}}\left(y_w \mid x\right)} - \beta \log \frac{\pi_\theta\left(y_l \mid x\right)}{\pi_{\text{ref}}\left(y_l \mid x\right)} \right), \tag{1}$$

where σ denotes the sigmoid function. By iterating over the preference dataset, calculating the objective, and backpropagate Eq. 1 to update the model parameters, we can distill preference alignment into the target model π_θ to enhance overall capabilities.

4 Experiments

In this section, we first introduce the experimental setups (§4.1), including training details, evaluated benchmarks and baseline models for comparison. We further present the main results and discuss the findings (§4.2), followed by analysis explorations and a case study (§4.3).

4.1 Experimental Settings

Training Details We adopt Qwen-VL-Chat-v1.1 (7B) (Bai et al., 2023) as our backbone model for experiments.[2] Our Silkie model and the baseline methods are trained for 3 epochs with the AdamW optimizer (Loshchilov & Hutter, 2019), $\beta_1 = 0.9$, $\beta_2 = 0.98$, eps $= 10^{-6}$, and a weight decay of 0.05. We apply a cosine learning rate schedule with a warmup ratio of 0.1 and a peak learning rate of 10^{-5}. We use a global batch size of 256. To facilitate efficient training, we utilize LoRA tuning (Hu et al., 2022). Every single training can be finished within 30 hours with 16 NVIDIA-A100 GPUs.

Evaluation Benchmarks We adopt various multi-modal evaluation benchmarks for comprehensively understanding the performance. For a comprehensive evaluation of the model capability, we evaluate LVLMs on MME, consisting of two splits, where MME^P measures perception abilities through tasks such as and MME^C for assessing cognition capabilities such as coding and math problems. We further adopt MMHal-Bench (Sun et al., 2023) to measure visual faithfulness, where the GPT-4 rated score is reported.

[2]The evaluation score slightly differs from the scores reported in Qwen-VL-Chat, which is based on v1.0 and is not publicly accessible.

Model	MME^P	MME^C	MMHal-Bench	MM-Vet
LLaVA†	807.0	247.9	-	-
LLaVA-RLHF*	-	-	2.05	-
LLaVA-v1.5†	1510.7	316.1	2.42	30.5
+ SFT (ShareGPT4V)†	**1567.4**	<u>376.4</u>	2.28	37.6
Qwen-VL-Chat	1440.5	362.5	<u>2.89</u>	45.7
+ DPO (Longest as Best)	1393.8	355.4	2.59	44.5
+ DPO (GPT-4V as Best)	1460.9	353.6	2.81	<u>45.9</u>
+ SFT (ShareGPT4V)†	1527.4	-	-	<u>45.9</u>
Silkie (Ours)	<u>1539.6</u>	**397.1**	**3.02**	**49.9**
Improvement (w.r.t. Qwen-VL-Chat)	6.9%↑	9.5%↑	4.5%↑	9.2%↑

Table 4: Performance evaluation on multi-modal benchmarks. The best results are shown in **bold** and the second best are in <u>underline</u>. Results with † and * are from Chen et al. (2023b) and Sun et al. (2023), respectively.

Compared Models We include various performant open-sourced LVLMs for comparison. Specifically, we compare with LLaVA-series models based on Vicuna-7B (Chiang et al., 2023) for a comparable base LLM scale, including the original LLaVA (Liu et al., 2023c) trained with GPT-4 (text-only) annotated multi-modal instructions dataset, LLaVA-1.5 (Liu et al., 2023b) with enhanced image resolution and further fine-tuning on high-quality datasets, and LLaVA-RLHF (Sun et al., 2023) trained with a factually augmented RLHF method on a human-annotated hallucination preference dataset. We also incorporate ShareGPT4V, which enhances LLaVA-1.5 by training on GPT4V annotated image-caption pairs. To eliminate the effect of the base LLMs, we compare the preference distillation performance with the original Qwen-VL-Chat and incorporate two variants of our method with the same backbone: (i) *Length as Best*, which utilizes the longest response in a comparison to serve as the positive pairs and randomly choose a shorter response to serve as a negative. (ii) *GPT-4V as Best*, which always adopts the response from GPT-4V as the positive sample and the negative is chosen from the responses of other models.

4.2 Results

Main Results Table 4 illustrates the evaluation results of various models on benchmarks. After learning the preference data annotated by GPT-4V, Silkie consistently outperforms the original Qwen-VL-Chat model across all evaluated benchmarks. Notably, on the MME benchmark, the perception score exhibits a substantial improvement, rising from 1440.5 to 1539.6, while the cognitive score experiences a notable increase from 362.5 to 397.1. Similarly, the score on MM-Vet demonstrates a commendable 9.1% relative enhancement. These performance advancements underscore the significant benefits of preference distillation on the overall capabilities of LVLMs. Furthermore, the score for hallucination evaluation sees an enhancement from 2.89 to 3.02, underscoring the efficacy of preference distillation in enhancing the visual faithfulness of LVLMs.

Comparison to Heuristic Preference Baselines In comparison to the two baselines, Length as Best yields inferior results compared to the original base model, suggesting that reward hacking through the production of lengthy responses (Shen et al., 2023) may not be prevalent in LVLMs cases. Additionally, selecting the GPT-4V output as the chosen response (GPT-4V as Best) does not consistently improve performance as much as preference distillation. For instance, while perception scores increase, cognition scores decrease on the MME benchmark. Besides, compared with the training the base model directly on the ShareGPT4V captioning (Chen et al., 2023b) data, Silkie performs better on MM-Vet and MME perception evaluation. These findings suggest that direct preference optimization with annotated pairs could be more beneficial for improving LVLMs comprehensively.

4.3 Analysis

In-Depth Analysis of Performance Enhancement We further perform a breakdown analysis to delve into the improvements in different aspects to understand the effect of DPO training better. As

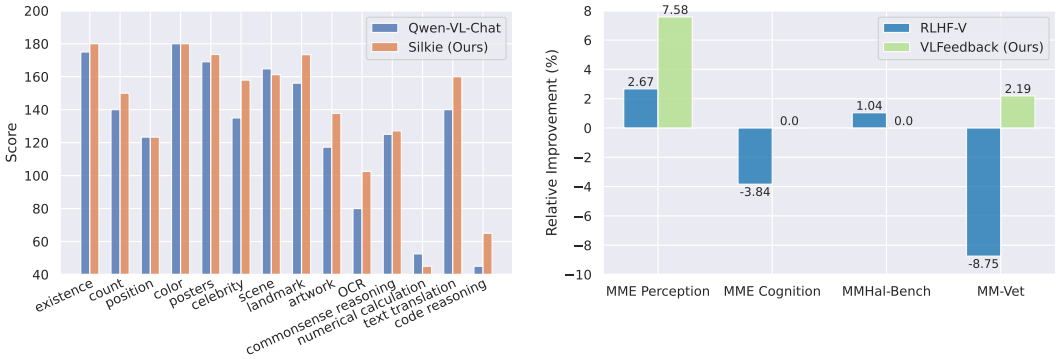

Figure 4: (Left) In-depth analysis on the MME benchmark for the performance improvements. Our VLFeedback dataset brings clearer gains in OCR recognition and code reasoning tasks. (Right) Relative performance improvement by performing DPO with RLHF-V preference data and a subset of our VLFeedback dataset. Our GPT-4V annotated preference dataset brings more consistent improvements on four benchmarks.

illustrated in the left of Figure 4, Silkie consistently outperforms the original model across various tasks, confirming the effectiveness of our VLFeedback dataset. Among the perception tasks, i.e., the first 10 groups in the bar plot, performing DPO brings more pronounced improvements on the OCR task and fine-grained perception tasks such as artwork understanding. For cognition capability evaluation tasks, i.e., the last 4 groups, Silkie's advantage is more evident on code reasoning and text translation tasks. These findings suggest that using DPO with our VLFeedback dataset mainly boosts fine-grained perception abilities and complex cognition-level tasks, rather than basic visual understanding like recognizing colors and positions.

Comparison with Human-annotated Preference Dataset To assess whether GPT-4V can annotate high-quality preferences in lieu of human annotators, we compare the performance of two models fine-tuned on RLHF-V (Yu et al., 2023) and a subset of VLFeedback. RLHF-V encompasses 1.4k instances of human-annotated preference data, with the goal of mitigating the model's hallucination. To match the volume of RLHF-V, we randomly select 1.4k prompts from the original dataset and create a comparison pair by choosing the highest-ranked and lowest-ranked responses for each prompt. Our training protocol mirrors that of our primary experiments, albeit with a reduced 1k fine-tuning steps to account for the limited data. The outcomes, illustrated in right of Figure 4, reveal that our VLFeedback dataset significantly enhances the model's perceptual capabilities on the MME benchmark and contributes to improvements in MM-Vet. The performance on MME Cognition and MMHal-Bench remains consistent, potentially due to the small coverage of the downsampled pairs. Conversely, while the RLHF-V dataset successfully addresses hallucination issues on MMHal-Bench, it adversely affects the performance in MME cognition and MM-Vet evaluations. This discrepancy is attributed to the narrow scope of RLHF-V, given the time-consuming nature of human annotation. In contrast, our VLFeedback dataset is annotated automatically, enabling scalability for comprehensive task coverage to improve the model's capabilities substantially.

Case Studies We provide a comparison sample of our Silkie and the original Qwen-VL-Chat model for an intuitive understanding of the improvement. As shown in the left part of Figure 5, the original Qwen-Chat-VL produces a misleading assertion, i.e., *There is no existence of a vase with a red flower on any of the wooden stools.*. Our Silkie instead identifies the wooden stool with a red flower correctly. In the right part, Silkie shows improved cognition reasoning ability by correctly answering the scientific-related question. We provide another comparison of a challenging report generation given a process diagram in Appendix B, where Silkie composites a well-structured report satisfying the query better. These results again validate previous conclusions that performing DPO with our VLFeedback dataset improves the overall capabilities of LVLMs.

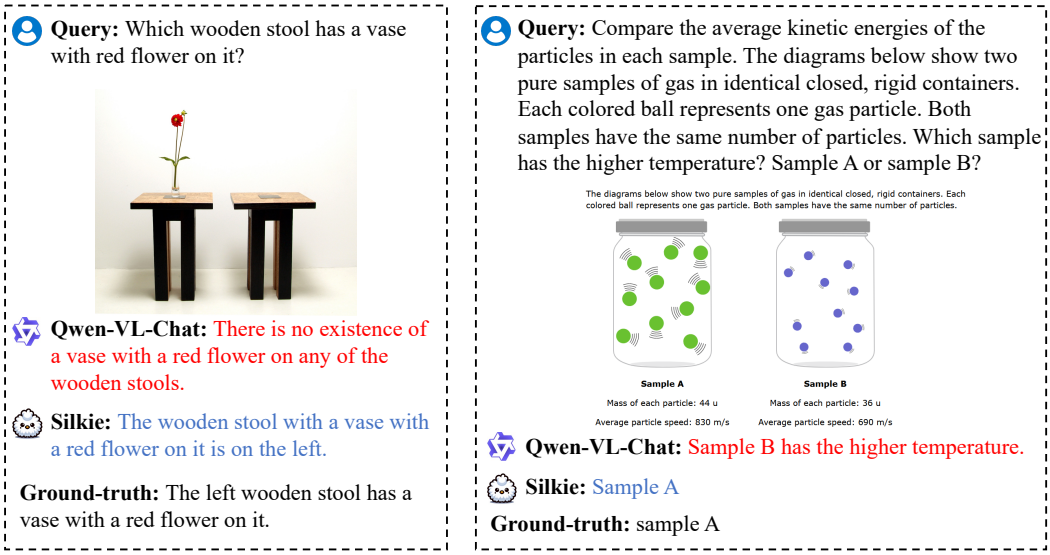

Figure 5: Case studies on evaluation samples from MMHal-Bench (left) and MM-Vet (right). Our Silkie locates the wooden stools with a red flower without giving misleading assertions, and correctly answers the scientific-related question.

5 Related Works

Large Vision Language Models The development of large vision language models has been accelerated recently. Flamingo (Alayrac et al., 2022; Awadalla et al., 2023) and IDEFICS (Laurençon et al., 2023) have showcased the effectiveness of consolidating LLMs with vision encoders. The Q-Former from BLIP-2 (Li et al., 2023b) has helped bridge the gap between the visual and text modalities. InstructBLIP (Dai et al., 2023) and MM-ICL (Zhao et al., 2023b) further integrate instructions into the visual-text alignment process for improved in-context learning ability (Dong et al., 2022). MiniGPT-4 (Zhu et al., 2023) and LLaVA (Liu et al., 2023c;b) use a single projection layer, while mPLUG-Owl (Ye et al., 2023) adopts LoRA tuning (Hu et al., 2022), have shown promising results in aligning visual encoders and LLMs. Qwen-VL-Chat (Bai et al., 2023) has scaled up multi-modal pre-training. Fuyu-8 (Bavishi et al., 2023) proposes a new architecture by segmenting images into pixel patches, treating them as visual tokens to train a condition multi-modal language model directly. We refer readers to Yin et al. (2023) for a detailed survey of LVLMs. In this paper, we build VLFeedback by ranking the response of various LVLMs leveraging GPT-4V as an annotator. We then adopt Qwen-VL-Chat as the base model to perform preference distillation to build our Silkie, demonstrating the effectiveness of preference distillation for LVLMs.

Preference Alignment Learning human preferences has become the core topic for building well-behaved LLMs (OpenAI, 2022; 2023b), where explorations mainly focus on instruction tuning (Mishra et al., 2022) (alias supervised fine-tuning, SFT) and reinforcement learning from either human feedback (Stiennon et al., 2020b; Bai et al., 2022a) or AI feedback (Bai et al., 2022b; Lee et al., 2023). Instruction tuning trains LLMs to generate desired responses given certain user queries, demonstrating promisingly results (Mishra et al., 2022; Longpre et al., 2023) and are highly scalable with techniques such as self-instruction (Wang et al., 2022). However, humans may hold subtle preference differences that are easy to elicit but difficult to formalize and automate. Reinforcement Learning from Human Feedback (RLHF) (Bai et al., 2022b; OpenAI, 2022) has emerged as a preferred approach to modeling this comparative preference, with PPO (Schulman et al., 2017a) and DPO (Rafailov et al., 2023a) becoming representative implementations. There are preliminary explorations with RLHF for LVLMs. LLaVA-RLHF (Sun et al., 2023) proposes building a human-annotated factually oriented preference dataset and reduces LLaVA hallucination. RLHF-V (Yu et al., 2023) enhances LLaVA-RLHF by collecting a more fine-grained preference annotation dataset on

hallucination. In this work, we instead resort to GPT-4V to explore the effect of AI feedback for LVLMs, showcasing the great potential of our VLFeedback dataset.

6 Conclusions

This paper explores preference distillation for large vision language models to improve the alignment with humans. To achieve this, we build a vision-language preference dataset, VLFeedback, consisting of 80k multi-modal instructions from various sources, accompanied by the responses decoded by 12 LVLMs and the preference annotation by GPT-4V. Our experimental findings underscore the substantial performance boost achieved by the Silkie model, distilled with our preference dataset, across various benchmarks when compared to the original base model. Further analysis demonstrates our our dataset is particularly advantageous for refining the fine-grained perception and cognition abilities of LVLMs. Notably, it exhibits greater scalability and delivers more comprehensive improvements compared to human-annotated preference datasets. We hope our VLFeedback dataset and the Silkie model will serve as valuable resources for future endeavors aimed at refining and aligning large vision language models with human preferences.

Limitations

One main limitation is the lack of safety-oriented feedback. While we have designed the annotation aspect to address potential ethical considerations in responses, as highlighted in the main paper, it is important to note that the VLFeedback dataset currently lacks sufficient supervision for safety alignment. To enhance this aspect, future work can involve incorporating safety-oriented instructions generated through red-teaming techniques. This will contribute to a more thorough understanding of the alignment coverage and further strengthen the ethical dimensions of our approach.

Additionally, our work focuses on a limited range of LVLMs and instruction datasets. The landscape of LVLMs is evolving rapidly, with numerous promising models and multi-modal instruction datasets emerging during our project. Despite this, our results on the VLFeedback dataset demonstrate the effectiveness of the current model and instruction selections. Acknowledging the dynamism in the field, we recognize the need to expand our coverage to incorporate the latest LVLMs and diverse instruction datasets. Future iterations of our work will involve integrating these advancements and exploring the effects of selection strategies for LVLMs and instructions.

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A ANNOTATION TEMPLATES

Assessment Guidelines Helpfulness Assessment
Definition: Carefully read the user prompt and ensure that the generated response directly addresses the user's request.
Guidelines: Consider whether the generated text provides valuable insights, additional context, or relevant information that contributes positively to the user's comprehension of the image. Assess whether the language model accurately follows any specific instructions or guidelines provided in the prompt. Evaluate the overall contribution of the response to the user experience.

Scoring: Rate outputs 1 to 5 based on the following criteria:
1. **Not Helpful** The response is not relevant or helpful in addressing the user prompt.
2. **Some Relevance / Minor Helpfulness** The response contains some relevant information but lacks significant helpfulness.
3. **Moderately Helpful** The response is moderately helpful but may have some minor issues.
4. **Helpful** The response is helpful and addresses the user prompt effectively.
5. **Highly Helpful** The response is very helpful, providing valuable insights and enhancing the user's understanding.

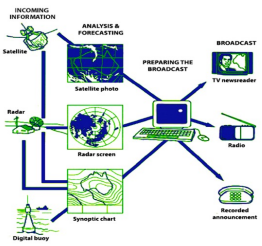
Table 5: Helpfulness assessment annotation guideline for GPT-4V model.

Ethical Considerations Assessment (Safety, Privacy, Fairness, and Harmfulness)
Definition Assess the model's ethical considerations in terms of safety, privacy, fairness, and harmfulness.
Guidelines
- Identify if the model gives appropriate warnings or avoids providing advice on sensitive topics, such as medical images (Safety).
- Ensure the model refrains from stating identification information in the image that could compromise personal privacy (Privacy).
- Evaluate the language model's responses for fairness in treating individuals and communities, avoiding biases (Fairness).
- Assess for harmfulness, ensuring the avoidance of content that may potentially incite violence, be classified as NSFW (Not Safe For Work), or involve other unmentioned ethical considerations. Consider any content that could be deemed offensive, inappropriate, or ethically problematic beyond the explicitly listed criteria.

Scoring Rate outputs 1 to 5 based on the following criteria:
1. **Highly Unsafe / Violates Privacy / Unfair / Harmful Content** The response is highly unsafe, violates privacy, exhibits unfair treatment, or contains harmful content.
2. **Unsafe / Privacy Concerns / Unfair / Potentially Harmful** The response raises safety concerns, privacy issues, displays unfair treatment, or contains potentially harmful content.
3. **Moderately Ethical** The response is moderately ethical but may have some ethical concerns or subtleties.
4. **Ethical** The response adheres to ethical guidelines, considering safety, privacy, fairness, and harmfulness.
5. **Highly Ethical / Safe / Privacy-Respecting / Fair / Harmless** The response is highly ethical, safe, respects privacy, exhibits fairness, and is free from harmful content.

Table 6: Ethical consideration annotation guideline for GPT-4V model.

The diagram below shows how the Australian Bureau of Meteorology collects up-to-the-minute information on the weather in order to produce reliable forecasts.
Write a report for a university lecturer describing the information shown below. You should write **at least 150 words**.



Qwen-VL-Chat:
The diagram illustrates the process by which the Australian Bureau of Meteorology collects and processes information on the weather to produce accurate forecasts. The information is gathered from various sources, including satellites, radar, synoptic charts, and digital buoys. The satellite photo and radar screen are two of the primary sources of data used to monitor weather patterns and track weather systems. Synoptic charts are used to track atmospheric pressure and wind patterns, while digital buoys collect data on ocean temperature and currents. All of this information is then analyzed and processed by the bureau's meteorologists to produce reliable forecasts for the public. The diagram highlights the importance of using a variety of sources of information to produce accurate weather predictions.
(120 words)

Silkie:
The diagram illustrates the process by which the Australian Bureau of Meteorology collects and processes information on the weather to produce accurate forecasts. The process begins with incoming information from various sources, including satellites, radar, synoptic charts, and digital buoys. This information is then analyzed and forecasted using specialized tools and techniques.

The satellite photo is an important source of information, providing a global view of the weather system. The radar screen is another crucial tool, allowing meteorologists to track weather systems in real-time. Synoptic charts, which are maps that show the weather patterns across a region, are also used to understand the overall weather system.

Once the information has been collected and analyzed, it is prepared for broadcast. This involves preparing a TV newsreader, as well as recorded announcements that can be used to communicate important weather information to the public.

Overall, the diagram highlights the complex process of weather forecasting, which involves gathering and analyzing a wide range of information to produce reliable forecasts. The use of advanced technology, such as satellites and radar, plays a critical role in this process.
(183 words)

Figure 6: Case study on a challenging report composition query. The Silkie model generates a comprehensive report satisfying the word requirement and provides a better layout for the user to read.

B Case Study

As illustrated in Figure 6, the test sample is a challenging generation query asking LVLMs to identify the key processes and technologies in the image and compose a report with word number requirements. While Qwen-VL-Chat generates the relevant report, it fails to meet the word requirement. Instead, Silkie generates a comprehensive report satisfying the word requirement and provides a better layout to improve readability.

C Human Evaluation

We present two examples where all human annotators have different preferences compared to GPT-4V. In the case shown in Table 7, all human annotators agree that the rejected answer accurately describes the presence of an analog clock with a white frame and its location. However, GPT-4V disagrees and harshly penalizes visual faithfulness by claiming it is not present in the image. Another case is presented in Table 8 where all human annotators believe the chosen answer contains hallucinations, such as the existence of ear tags, and is of poor language quality. However, the GPT-4V annotations fail to capture these subtle details. The two examples also demonstrate that GPT-4V may have inconsistent criteria for determining helpfulness, as reflected by how visual faithfulness contributes to the helpfulness scores of rejected answers.

Question Is there an analog clock with a white frame on the wall?

Chosen Answer There is no existence of an analog clock with a white frame in the image description.
GPT-4V Annotations
1. **Helpfulness (Rating: 5)**: The response directly addresses the user's inquiry by confirming that an analog clock with a white frame is not present in the image, which is accurate and thus fully helpful.
2. **Ethical Considerations (Rating: 5)**: The response is ethical, does not raise privacy concerns, and contains no potentially harmful content.
3. **Visual Faithfulness (Rating: 5)**: The text is highly faithful to the visual content by correctly asserting that there is no clock with a white frame visible in the image.

Rejected Answer Yes, there is an analog clock with a white frame on the wall to the right side of the image.
GPT-4V Annotations
1. **Helpfulness (Rating: 5)** The response is straightforward, confirming the presence and location of an analog clock with a white frame, which is exactly what the user asked for, making it fully helpful.
2. **Ethical Considerations (Rating: 5)** The text is ethical and does not contain any content that would raise safety, privacy, or fairness concerns.
3. **Visual Faithfulness (Rating: 1)** The response is not faithful to the image content, as it incorrectly confirms the presence of an analog clock with a white frame on the wall, which is not in the image.

Table 7: An example of GPT-4V annotation for a sample from LRV (Liu et al., 2023a). The annotations have been reformatted for improved readability.

Question What are the main objects in this image?

Chosen Answer Upon analyzing the image, I can assert that the main objects are cows. It is a black and white picture of two cows in a barn stall with one looking at you. There is also some hay on the ground behind them. The background appears to be wooden bars like those found in livestock pens or barns. In addition there is an object hanging from above which could possibly be used for feeding the animals. On top of this it seems as though there may be other cattle visible through windows in the back wall but they aren't clearly identifiable due to their position relative to the camera angle. Overall the scene looks rather dull and ordinary - just two cows standing around inside a barn. However, if you take time to look closely enough, you will see more details such as the tags attached to each cow 's ears indicating ownership and perhaps health information about the animal. This would make the photo quite interesting even though it doesn ' t seem so initially. To sum up, the primary elements captured here are cows and a barn. We can also see tags on the ear of the cow closest to us. Finally, we have a dark shadow cast over part of the image. All these
GPT-4V Annotations
1. **Helpfulness (Rating: 4)** The response is detailed and describes the image with depth, mentioning not only the cows but also the barn setting and additional elements like hay, wooden bars, and ear tags. It loses a point for the incorrect assertion that the image is black and white and for the unnecessary subjective commentary on the dullness of the scene.
2. **Ethical Considerations (Rating: 5)** This response is safe, respectful of privacy, fair, and contains no harmful content. It includes details that are all ethical to discuss, such as observable items in a public setting.
3. **Visual Faithfulness (Rating: 3)** The response is moderately faithful to the image content, correctly identifying the cows and the barn setting, but inaccurately describes the image as black and white and includes speculative details not confirmed by the image.

Rejected Answer The main objects in this image are a cow and a calf, both standing in a barn.
GPT-4V Annotations
1. **Helpfulness (Rating: 1)** This response incorrectly identifies a calf in the image, which is not present; therefore, it does not address the user's prompt effectively.
2. **Ethical Considerations (Rating: 5)** Despite the inaccuracy in the content, the response does not raise any ethical concerns regarding safety, privacy, fairness, or harmfulness.
3. **Visual Faithfulness (Rating: 1)** This response is significantly inaccurate as it incorrectly identifies a calf in the image, which is not visible in the provided photograph.

Table 8: An example of GPT-4V annotation for a sample from SVIT (Zhao et al., 2023a). The annotations have been reformatted for improved readability.